

	STUDENT NAME			COURSE NAME	Certified in MEP (HVAC)
	REGISTER. NO			TOTAL HOURS	90 HRS
	MOBILE.NO			JOINING DATE	
	CLASS TIMING			STAFF NAME	

SESSION	DATE & HOURS	MODULES	HOURS	STUDENT SIGN	STAFF SIGN
01 SESSION		AUTOCAD MEP			
	DATE : _____ IN TIME : _____ OUT TIME : _____	INTRODUCTION TO ENGINEERING DRAWINGS • Introduction to CAD • Projections ➤ Advantages • Applications of CAD • File Management • New, Open • Save, Save as • Close, Exit • Getting Started with AUTOCAD • Units, Limits • Dimensions Style • Drawing Simple Sketches • Line, Circle, Arc • Ellipse, Elliptical arc • Rectangle, Polygon • Function keys and Uses • Editing Commands • Erase, Oops, Undo, Redo • Move, Copy, Rotate, Array • Mirror, Fillet, Chamfer • Scale, Stretch, Offset, • Trim, Extend • Dimension Measurements • Linear, Aligned, Angle • Radius, Diameter • Arc length, Area • ANNOTATIONS • Text • Single Line Text (DTEXT) • Multi Line Text (MTEXT) • Text edit, Text Style (ST) • Leaders, Multi leader, Leader Align •	180 Mins		

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SESSION	DATE & HOURS	MODULES	HOURS	STUDENT SIGN	STAFF SIGN
02 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	2D Sketches • Draw 2D Sketches	240 Mins		
03 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	ISOMETRIC VIEW • Purpose of Isometric View Setting workspace for Isometric Drawing • Isocircle, Isoplane changing • Dimensioning for Isometric Drawing	120 Mins		
04 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	ISOMETRIC DRAWING • Draw Isometric Drawings	180 Mins		
05 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	LAYER PROPERTIES AND PAGE SETUP • LAYER • Layer Management • Application of Layer • Adding/Removing layers • Layer status, new property filter • New group filter • Layer status manager • INQUIRY COMMANDS • Groups • Uses of groups, create group, ungroup, edit group • Blocks • Usage of blocks, Insert block, Block editor. • PAGE SETTING & PLOTTING • Introduction to Plotting • Page Setup • Plot Styles, Plot	240 Mins		

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06 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	HVAC - 2D DESIGN - MEP INTRODUCTION TO HVAC • THEORY • Basic Introduction • Functions of HVAC • Unit conversion • Standards of HVAC • Heat & It Types, Mode of heat transfer	120 Mins		
07 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	REFRIGERANT CYCLE & PSYCHOMETRIC CHART • THEORY • REFRIGERANT CYCLE • Vapour compression cycle • Vapour absorption cycle • Components of Refrigerant cycle • (Compressor, Condensor, Expansion valve & Evaporator) • Filter & It Types • PSYCHOMETRIC CHART • Air Properties	90 Mins		
08 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	REFRIGERANTS & CLASSIFICATIONS OF AC'S • THEORY • REFRIGERANTS • Refrigerants Class & It Uses • CLASSIFICATIONS OF AC'S • Unitary AC & It Types • Centralized AC & It Types • AIR HANDLING UNITS	120 Mins		
09 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	DUCT DESIGN METHOD • DUCT DESIGN METHOD • Equal friction method • Velocity reduction method	180 Mins		

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10 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	AIR DISTRIBUTION SYSTEM • THEORY • AIR DISTRIBUTION SYSTEM • Duct & It basic theory • Air terminal units • SMACNA Standards	120 Mins		
11 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	DUCT DRAFTING	300 Mins		
12 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	DUCT SIZER SOFTWARE • DUCT SIZE CALCULATION • SPLIT DUCT DRAFTING • FLEXIABLE DUCT DRAFTING	150 Mins		
13 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	VENTILATION SYSTEM • THEORY & PROBLEMS SOLVING • VENTILATION SYSTEM • Carparking • Toilet • Kitchen hood	180 Mins		
14 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	VETILATIONS DRAFTINGS • Drafting • KITCHEN HOOD VENTILATION • CARPARKING VENTILATION • TOILET VENTILATION	300 Mins		

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15 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	AIR TERMINAL SIZE CALCULATION AIR TERMINAL SIZE CALCULATION USING BETA ATU AND AUTOCAD SOFTWARE	120 Mins		
16 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	HEAT LOAD CALCULATION - Formulae - Problems For load calculation	240 Mins		
17 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	HEAT LOAD CALCULATION ASSESSMENT PROBLEMS SOLVING	180 Mins		
18 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	HEAT LOAD CALCULATION USING SOFTWARE HEAT LOAD CALCULATION (Hourly Analysis Program)	180 Mins		
19 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	CHILLER & VRF/VRV - THEORY - CHILLER & VRF/VRV - Types of chillers - Working principle - Refrigerant piping & fitting selection	180 Mins		

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20 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Chiller pump calculation Chiller pump calculation	180 Mins		
21 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	HVAC PROJECT HVAC PROJECT : NEED AUTOCAD AND PDF FILES	600 Mins		
22 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	REVIT MEP - 3D MODELLING - MEP Introduction - Introduction to BIM - Start a New project - Introduction to revit user Interface - Selection in Revit and Modify tools	120 Mins		
23 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Project Setup - Revit Coordinates System - Link Architectural Model and manage it - Setting up Grids and Levels - Create Plans, Link design models and view range - Visibility Graphics, Parameters, filters and View - templates - Scope box and Sections box	120 Mins		
24 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Spaces and Analysis - Create and Modify Spaces - Energy Analysis(Heat Load Calculation)	120 Mins		

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25 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	HVAC 3D model Creation • Introduction to HVAC system • Place and modify Mechanical Equipments • Place and Modify Duct and Duct fitting • Place and Modify Air terminals and Duct accessories • Place and Modify Duct Insulation • Place and Modify HVAC pipes and pipe accessories	180 Mins		
26 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Plumbing 3D model Creation • Introduction to Plumbing system • Place and modify Plumbing Fixtures & Equipments • Place and modify Drainage pipe and pipe accessories • Place and modify Water supply pipe and pipe accessories.	180 Mins		
27 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Fire Fighting 3D model Creation • Introduction to Fire Fighting system • Place and modify Fire Fighting Equipments • Place and modify Fire Fighting pipe and pipe accessories • Place and modify Sprinklers	180 Mins		
28 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Electrical 3D model Creation • Introduction to Electrical system • Place and modify Lighting Fixtures • Place and modify power terminals and panel boards • Place and modify Data Device • Create Circuits • Place and modify Cable tray and Cable tray fittings • Place and modify Conduits and Conduits fittings	180 Mins		

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29 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Schedules and Material Takeoff Report • Creating and Modifying HVAC Schedules • Creating and Modifying Pipe Schedules • Creating and Modifying Electrical Schedules	120 Mins		
30 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Design Drawing Extraction • Choose and Create Right Title Block • Add Views into the Sheet • Annotation and Taggings • Shop Drawing Extraction	120 Mins		
31 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	FINAL PROJECT • Final Project Exercises	720 Mins		

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32 SESSION	DATE : _____	NAVIS WORK (MEP) Introduction - Navisworks product family (Manage, Simulate, Freedom). - Interface overview: Ribbon, Navigation bar, Selection tree. - File formats (NWF, NWD, NWC).	30 Mins		
	IN TIME : _____				
	OUT TIME : _____				
33 SESSION	DATE : _____	File Handling & Integration - Importing CAD/Revit/IFC models. - Append vs Merge. - Managing coordinates and units.	60 Mins		
	IN TIME : _____				
	OUT TIME : _____				
34 SESSION	DATE : _____	Navigation Tools - Walk, Fly, Orbit, Zoom, Pan. Steering wheel, View cube. - Saving viewpoints.	60 Mins		
	IN TIME : _____				
	OUT TIME : _____				

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35 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Selection & Visibility Control - Selection sets, search sets. Hide/Unhide, override materials, - transparency. Section planes.	60 Mins		
36 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Clash Detection Basics Setting up clash tests, rules, clash matrix.	60 Mins		
37 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Clash Detection Advanced Grouping clashes, filtering, generating reports (HTML, Excel).	60 Mins		
38 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	TimeLiner Basics - Linking schedules (MS Project/Excel). Creating tasks, - assigning resources.	60 Mins		
39 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	TimeLiner Advanced Running simulations, exporting animations, project tracking.	60 Mins		

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40 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Quantification Basics • Item catalog setup, manual takeoffs	45 Mins		
41 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Quantification Advanced • Automated takeoffs, Takeoff palette, exporting quantities.	45 Mins		
42 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Animator Basics • Object animations, camera paths	45 Mins		
43 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Scripter Advanced • Event triggers, linking animations with TimeLiner.	45 Mins		
44 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Rendering Basics • Applying materials, lighting setup.	30 Mins		

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45 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Rendering Advanced Exporting high-quality images/videos.	30 Mins		
46 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Collaboration Tools Redline markup, comments, viewpoints with notes. Publishing NWD files. Using Freedom viewer	60 Mins		
47 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Final Project Integrated project: Clash detection, rendering, collaboration. Deliverables: Report, video, cost sheet, walkthrough.	240 Mins		

COURSE **COMPLETED**

Staff sign

Student sign